

# CIRCLE PROPERTIES

## Foundation Tier

Circle vocabulary and prerequisite geometry

Name .....

Date .....

7 questions

15 marks

2017-2025

### Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

### Using this booklet

Questions are arranged by exam series. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

This booklet covers circle properties and prerequisite geometry. Standard circle theorems are Higher-tier content. Only the specified parts of each question are included.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers 2017-2025. Questions are included only in the booklet that matches their main purpose.

# Question index

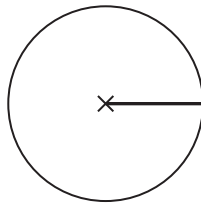
Foundation Tier | 7 questions | 15 marks

| Practice | Exam series   | Paper   | Original question | Marks | Page |
|----------|---------------|---------|-------------------|-------|------|
| 01       | June 2018     | 1MA1/1F | Q9                | 2     | 3    |
| 02       | November 2019 | 1MA1/3F | Q10               | 2     | 4    |
| 03       | June 2022     | 1MA1/2F | Q8(c)             | 2     | 5    |
| 04       | June 2023     | 1MA1/1F | Q6                | 1     | 6    |
| 05       | November 2023 | 1MA1/3F | Q29               | 4     | 7    |
| 06       | June 2024     | 1MA1/3F | Q10               | 2     | 8    |
| 07       | November 2025 | 1MA1/1F | Q9(a)-(b)         | 2     | 9    |

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

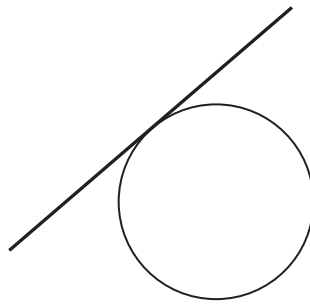
- 9 The centre of this circle is marked with a cross (×).



- (a) Write down the mathematical name of the straight line shown in the circle.

.....  
(1)

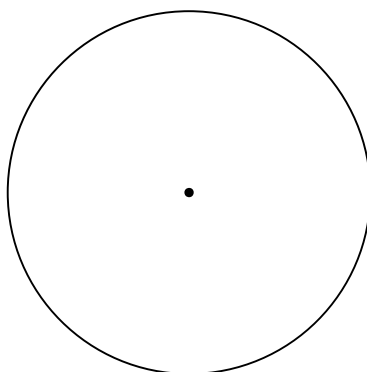
- (b) Write down the mathematical name of the straight line that is touching the circle.



.....  
(1)

**(Total for Question 9 is 2 marks)**

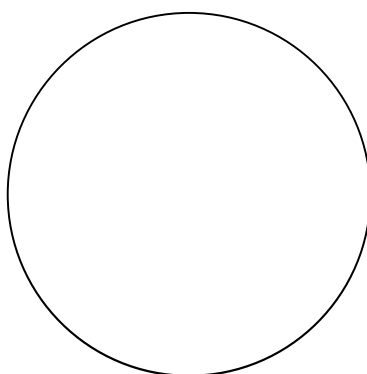
Additional working space

**10**

- (a) On the diagram above, draw a diameter of the circle.

(1)

- (b) On the diagram below, draw a segment of the circle.  
Shade the segment.



(1)

---

**(Total for Question 10 is 2 marks)**

---

Additional working space

---

---

---

---

---

---

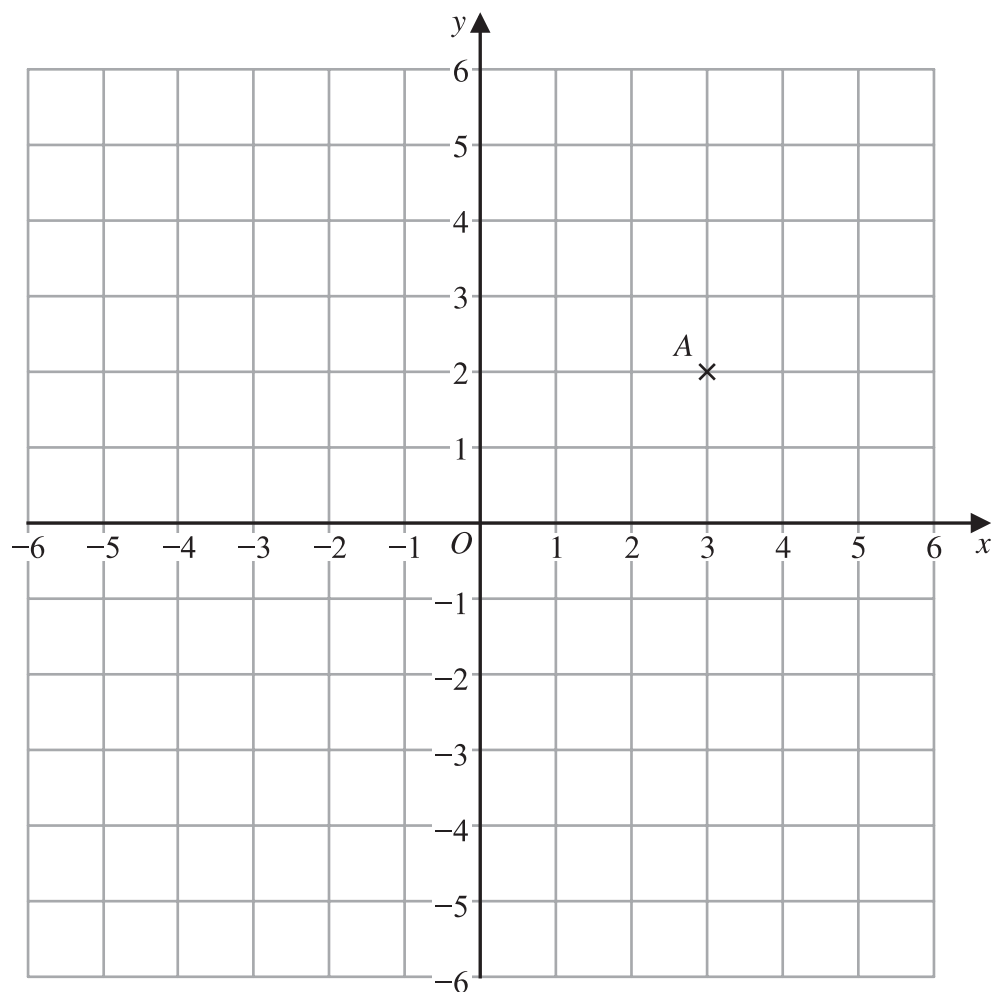
---

---

---

---

8 Here is a centimetre grid.



- (c) On the grid, draw the circle with  
centre  $(1, -1)$   
and radius 4 cm.

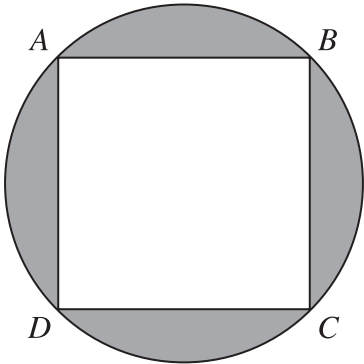
(2)

Additional working space

A diagram of a circle with a center point. A line segment passes through the center, with arrows at both ends pointing to the circle's boundary, labeled "Diameter". An arrow points to the outer edge of the circle, labeled "Circumference". An arrow points to the center point, labeled "Centre".

### Additional working space

29  $A$ ,  $B$ ,  $C$  and  $D$  are points on a circle such that  $ABCD$  is a square.



The square  $ABCD$  has sides of length 3.5 cm.

Calculate the circumference of the circle.  
Give your answer correct to 1 decimal place.  
You must show all your working.

..... cm

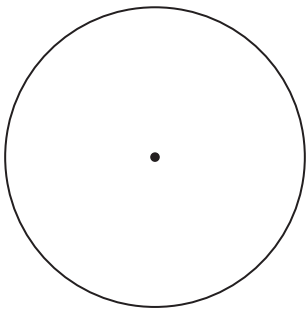
(Total for Question 29 is 4 marks)

Additional working space

.....

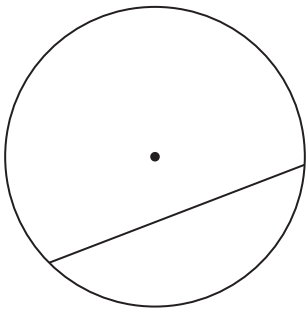
.....

10 Here is a circle.



- (a) On the diagram above, draw a radius of the circle.
- (1)

Here is another circle.



- (b) Write down the mathematical name for the straight line inside this circle.

.....

(1)

(Total for Question 10 is 2 marks)

Additional working space

.....

.....

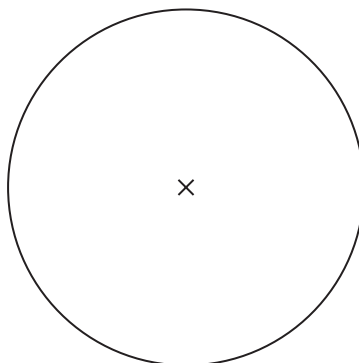
.....

.....

.....

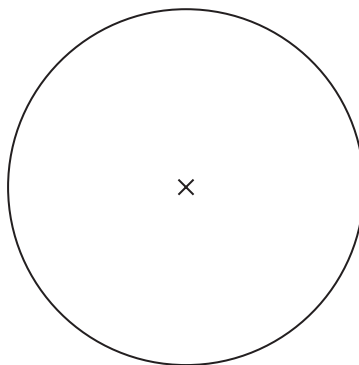
.....

- 9 (a) On the circle below, draw a radius.



(1)

- (b) On the circle below, draw a chord.



(1)

Additional working space

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....